

# Cell Design Specification — VBF-T2R1FLASH-DS-01

**\*\*Case\*\*:** t2\_r1\_flash · Grid energy storage cell · Generation date: 2026-08-25

**\*\*Numbering\*\*:** VBF-T2R1FLASH-DS-01

## 1. Basic specification

| Item                         | Value                                                                                                                         | Source                                              |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| Electrochemical system       | NMC811 / graphite (Chen2020 parameter set)                                                                                    | entry 0 meta.base                                   |
| Nominal capacity             | 5.0 Ah (parameter set); 1C delivered 5.0469 Ah                                                                                | parameter set / cell/r2_fcc_1c_dfn.json:capacity_ah |
| Voltage window               | 2.5 – 4.2 V                                                                                                                   | parameter set cut-offs                              |
| Cell dimensions (h x w x t)  | 65.0 x 1580 x 0.2008 mm (electrode stack, no casing)                                                                          | parameter set / calc-energy thickness_m             |
| Electrolyte formulation      | Advanced LiFSI-class high-transport: sigma 2.0 S/m, D 2.5e-10 m2/s, t+ 0.55 (transport overrides, literature-order estimates) | params/fcc.json, propose r2c                        |
| Cation transference number   | 0.55 (override; baseline 0.2594)                                                                                              | params/fcc.json                                     |
| Cooling (thermal management) | Total heat transfer coefficient 60 W/m2K (active liquid cooling, design choice)                                               | params/fcc.json                                     |

## 2. Electrode and separator

| Item                                    | Value                                                                                                                                                                     | Source                          |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| Positive electrode thickness / porosity | 75.6 um / 0.335                                                                                                                                                           | parameter set                   |
| Negative electrode thickness / porosity | 85.2 um / 0.30 (porosity override 0.25->0.30)                                                                                                                             | parameter set / params/fcc.json |
| Separator thickness / porosity          | 12 um / 0.47                                                                                                                                                              | parameter set                   |
| Positive current collector              | Al 16 um                                                                                                                                                                  | parameter set                   |
| Negative current collector              | Cu 12 um                                                                                                                                                                  | parameter set                   |
| Positive particle radius                | 4.0 um (override 5.22->4.0 um)                                                                                                                                            | params/fcc.json                 |
| Negative particle radius                | 2.5 um (override 5.86->2.5 um)                                                                                                                                            | params/fcc.json                 |
| N/P ratio                               | 0.97 (density caliber: rho x (1-eps) x L x usable stoich; neg dx 0.05-0.90, pos dy 0.27-0.99; c_max-caliber inconsistent with densities in this parameter set, annotated) | computed, formula in calc.xlsx  |

## 3. Process design parameters

| Parameter                   | Formula                            | Value | Unit                                                                   |
|-----------------------------|------------------------------------|-------|------------------------------------------------------------------------|
| Positive areal density      | thickness x (1-porosity) x density | 0     | g/m2                                                                   |
| Negative areal density      | thickness x (1-porosity) x density | 0     | g/m2                                                                   |
| Positive compaction density | density x (1-porosity) / 1000      | 2.17  | g/cm3                                                                  |
| Negative compaction density | density x (1-porosity) / 1000      | 1.16  | g/cm3                                                                  |
| Electrolyte fill amount     | pore volume x 1.2 g/cm3 (lit.)     | 6.97  | g                                                                      |
| Formation recommendation    | 0.1C CC to 4.2 V, 25 C, 2 cycles   | —     | design recommended value; actual production-line value requires tuning |

## 4. Mass breakdown (formula caliber: layer thickness x area x (1-porosity) x density; electrolyte excluded per contract, included in BOM separately)

| Layer                                           | Mass (g)         | Source                         |
|-------------------------------------------------|------------------|--------------------------------|
| Positive electrode                              | 16.84            | calc-energy layer_kg_m2 x area |
| Negative electrode                              | 10.15            | calc-energy                    |
| Positive CC (Al)                                | 4.44             | calc-energy                    |
| Negative CC (Cu)                                | 11.04            | calc-energy                    |
| Separator                                       | 0.26             | calc-energy                    |
| <b>**Total (electrolyte-excluded caliber)**</b> | <b>**42.73**</b> | calc-energy mass_kg            |

## 5. Performance verification (vs entry-0 criteria)

| Metric                    | Value                                      | Threshold       | Verdict | Source                    |
|---------------------------|--------------------------------------------|-----------------|---------|---------------------------|
| Energy density            | 425.41 Wh/kg                               | $\geq 327.18$   | PASS    | cell/r2_fcc_energy.json   |
| SEI thickness @100 cyc 1C | 78.2 nm                                    | $\leq 500$      | PASS    | cell/r2_fcc_aging100.json |
| SEI thickness @500 cyc 1C | 262.0 nm                                   | $\leq 550$      | PASS    | cell/r2_fcc_derived.json  |
| -20C discharge retention  | 99.4 %                                     | $\geq 90$       | PASS    | cell/r2_fcc_derived.json  |
| 4C charge plating         | anode min 12.3 mV $\geq 0$ ,<br>no plating | no plating      | PASS    | cell/r2_fcc_4c.json       |
| 4C charge T_max (45C amb) | 329.2 K                                    | $\leq 333.15$ K | PASS    | cell/r2_fcc_4c.json       |

## 6. Design notes (changes vs baseline Chen2020 and why — traceable to evaluate log)

- Negative particle 5.86  $\rightarrow$  2.5  $\mu\text{m}$ , positive 5.22  $\rightarrow$  4.0  $\mu\text{m}$** : lower solid-diffusion overpotential  $\rightarrow$  lifts anode surface potential at 4C (plating); lower polarization/heat (R2 FC-A  $\rightarrow$  FC-C).
- Negative porosity 0.25  $\rightarrow$  0.30**: shorter effective electrolyte path in the negative (plating margin).
- Electrolyte transport overrides ( $\sigma$  0.95  $\rightarrow$  2.0 S/m,  $D$  1.77e-10  $\rightarrow$  2.5e-10 m<sup>2</sup>/s,  $t^+$  0.26  $\rightarrow$  0.55)**: high-transport LiFSI-class formulation; reduces ohmic + concentration polarization at 4C (estimates, marked in propose r2c).
- SEI kinetic rate constant 1e-12  $\rightarrow$  2e-15 m/s**: artificial-SEI/coating-class suppression bridge (estimate); drives SEI 449 $\rightarrow$ 78 nm @100 cyc and 778 $\rightarrow$ 262 nm @500 cyc (R2 scans showed  $k$  is the controlling lever:  $k \times 0.2 \rightarrow$  744 nm,  $k \times 0.01 \rightarrow$  583 nm).
- Cooling  $h$  10  $\rightarrow$  60 W/m<sup>2</sup>K**: active liquid cooling for grid storage; T\_max 4C 354  $\rightarrow$  329 K.
- Reverted negative thickness to 85.2  $\mu\text{m}$**  (FC-B experiment: 100  $\mu\text{m}$  negative worsened 4C anode dip -0.125 V via longer electrolyte path).
- Capacity-trajectory artifact in aging (climb-then-saturate under standard SEI model) is labeled honestly; SEI thickness is the reliable aging indicator.